Aadesh Shah

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EDUCATION

North Carolina State University Expected 12/2027 | Raleigh, NC
BS, Computer Engineering | GPA: 4.0/4.0
Relevant Coursework: Data Structures and Algorithms, Advanced Microprocessor Architecture, Parallel Architecture, Operating Systems, Linear Algebra, Embedded Systems, Digital ASIC/FPGA Design

EXPERIENCE

Software Development Engineer Intern 09/2026 – 11/2026 | Arlington, VA
Amazon Web Services (AWS)

- Developed a **full-stack agentic application** that turns natural-language requests into data center simulation inputs, using Amazon Bedrock AgentCore and the Strands Agents SDK with a web UI and CI/CD pipeline. Cut setup time from **hours to minutes** for routine changes and from **days to ~30 minutes** for complex scenarios.
- Built an automated ontology generator on AWS Fargate to derive schemas and a dependency graph from **80+ JSON configuration files**, enabling the agent to plan cross-file edits and validate them with Pydantic before user approval.
- Automated S3 cleanup of incomplete multipart uploads after finding ~**$12K** in accumulated charges, preventing ~**$800/month** in recurring storage costs.

Software Engineering Intern 06/2026 – 08/2026 | Durham, NC
Tanium

- Built a Go pre-scan filtering service for a vulnerability scanner, excluding ~**40%** of incoming Docker images as unprocessable and reducing Lambda failure rates from **30% to 1%**.
- Engineered a Python service aggregating vulnerability scan results across **40 production targets**, integrating CloudWatch and Slack APIs to surface critical failures in real time.
- Configured Grafana to detect failures and send alerts across **10 production security services**, resolving IAM barriers to cross-region monitoring.

Applied ML Research Assistant 08/2024 – Present | Raleigh, NC
North Carolina State University

- Designed and trained a **PyTorch** CNN for cough detection from wearable sensor data, achieving **98% validation accuracy**. Co-authored a paper on this work in *Electronics*.
- Quantized the model to ~**25% of its original size** and wrote a memory-efficient C pipeline to generate three-channel mel spectrograms. Deployed both on the MAX78000 MCU, achieving **15–16 ms latency** at ~**20 µJ per inference**.

R&D Embedded Software Intern 06/2025 – 08/2025 | Charlotte, NC
Columbus McKinnon

- Led C/C++ development of a control-system prototype for **5+ product lines**, incorporating direct customer input.
- Scheduled concurrent tasks with **FreeRTOS** and benchmarked three embedded UI frameworks on the **NXP i.MX RT1050**.

PROJECTS

Speculative Value Prediction Engine 01/2026 – 04/2026

- Extended a large-scale C++ **RISC-V** simulator with a novel hybrid value predictor, increasing throughput by ~**24%** across 15 benchmarks within a **32 KB** budget and placing **2nd out of 60 teams** in a graduate-level competition.
- Implemented the full prediction pipeline from scratch, including training, speculative injection, and misprediction recovery, drawing on techniques from recent computer architecture research.

AI-Powered C++ Code Review & Optimization Assistant 09/2025 – 10/2025

- Designed a C++ optimization pipeline using **Clang AST** analysis and an LLM feedback loop to propose source rewrites, achieving a **70%** mean runtime reduction across 22 custom benchmarks against Clang -O3 baselines.
- Developed a FastAPI/React application supporting multi-file project uploads, isolated Docker execution, optional Callgrind profiling, and optimized-source downloads with correctness checks, benchmark reports, and source diffs.

Stock Trading Application 06/2025 – 07/2025

- Created a multithreaded C++/MySQL trading platform for **low-latency order execution and management**.
- Integrated Twitter and news APIs for market sentiment, using API caching to reduce lookup latency by ~**80%** under concurrency; fine-tuned **FinBERT** for financial sentiment classification, achieving **88% accuracy**.

SKILLS

Languages: C++, C, Go, Python, Java, Verilog, SQL
Software Engineering & Systems: Concurrency, Multithreading, CUDA, OpenMP, REST APIs, CI/CD, Linux, LLVM/Clang
Machine Learning & AI: PyTorch, LLM Agents, Agent Evaluation, Transformers (BERT), CNNs, Model Quantization
Tools & Frameworks: Git, Docker, AWS, Amazon Bedrock AgentCore, Strands Agents SDK, FastAPI, React, Vivado